Methods & Reproducibility Log

Remote sensing and geospatial approaches for multi-hazard monitoring in Nepal

Systematic / hybrid review. Databases: Scopus + Web of Science (2016-2025). Screening: full manual title/abstract review.

1. Search strategy

Two databases - Scopus and the Web of Science Core Collection - were searched on 14 June 2026 (window 2016-2025) using five complementary Boolean streams (Table 1): a broad multi-hazard query and four hazard-specific queries that add hazard-appropriate sensor/method terms. The field operator was TITLE-ABS-KEY in Scopus and TS (Topic) in Web of Science.

Table 1. The five search streams.

Stream	Boolean query (Scopus TITLE-ABS-KEY; WoS TS)
Main	("remote sensing" OR satellite OR "earth observation" OR geospatial OR GIS) AND (flood* OR landslide* OR "glacial lake outburst flood" OR GLOF OR drought* OR "multi-hazard") AND Nepal
Flood	("remote sensing" OR satellite OR GIS OR "earth observation") AND (flood* OR inundation) AND Nepal
Landslide	("remote sensing" OR satellite OR GIS OR DEM OR InSAR) AND (landslide* OR "slope failure") AND Nepal
GLOF	("remote sensing" OR satellite OR GIS OR DEM) AND ("glacial lake outburst flood" OR GLOF OR glacier*) AND Nepal
Drought	("remote sensing" OR satellite OR GIS OR NDVI) AND (drought* OR "vegetation stress") AND Nepal

2. Eligibility

Inclusion required all three of: (i) Nepal as the actual study area; (ii) a concrete remote-sensing/geospatial method; (iii) a hazard-monitoring focus. Reviews, books and conference papers were retained separately for context, not as primary studies.

3. De-duplication

Records were pooled and de-duplicated by a logged routine using ordered cross-database keys: DOI, then title+year, then native id. Of 1,362 records, 855 were duplicates (843 DOI; 12 title+year), leaving 507 unique; 248 were indexed in both databases. Removing 24 non-article types left 483 articles for screening.

4. Screening (manual)

All 483 articles underwent manual title/abstract screening. A documented relevance classifier provided an initial sort and provisional tags, but every candidate was read and adjudicated against the eligibility test. Screening excluded 228 articles - 160 not a Nepal study area (mostly other High Mountain Asia regions, or generic methods not applied to Nepal), 52 off-topic (air quality, ecology/wildlife, social/livelihood, agronomy, governance), and 16 not remote-sensing-based - leaving 255 confirmed in scope. One further eligible article, identified through citation snowballing, was added on the same basis, for 256 in scope.

5. PRISMA record flow

Stage	n	Cumulative
Records identified - Scopus	567	
Records identified - Web of Science	795	
Total identified	1,362	1,362
Duplicates removed (DOI 843; title+year 12)	-855	507
Non-article doc types removed	-24	483
Excluded at title/abstract screening (manual)	-228	255
Added via citation snowballing	+1	256
Articles in scope	256	256
In-depth synthesis core (stratified)	100	100

6. Prioritisation of the synthesis core

From 256 in-scope articles, a core of 100 was selected by an explicit priority score (multi-hazard relevance 25%, thematic centrality 25%, operational content 15%, recency 15%, cross-database validation 10%, citations/yr 10%), stratified by hazard with a floor of ten per stratum. Core composition: Landslide 39, Multi-hazard 23, GLOF 16, Flood 15, Drought 7 (all available Nepal drought studies); 80 of 100 are from 2020 or later. A final manual QA pass replaced five weaker fits (generic deep-learning architectures and pure glacier-process studies) with Nepal-specific hazard studies of the same hazard.

7. Bibliometric snapshot (in-scope, n=256)

7.1 Publications by year

Year	n	Year	n
2016	19	2022	24
2017	29	2023	25
2018	23	2024	29
2019	24	2025	32
2020	22	2026	1
2021	28		

7.2 Database coverage

Database coverage	In-scope articles
Both databases	178
Web of Science only	48
Neither (added via citation snowballing)	1
Scopus only	29

7.3 By hazard

Hazard (primary)	In-scope articles
Landslide	113
Multi-hazard / cascade	69
GLOF / glacier	34
Flood	33
Drought	7

8. Reproducibility

The pipeline (merge_dedupe.py, classify_screen.py, manual decisions, rank_priority.py) and all derived datasets are provided as supplementary material and in a public repository. Raw Scopus/WoS records are not redistributed (database licensing); the repository shares scripts, DOI-keyed decisions, and instructions to regenerate the corpus.